Department of
Computer Science and Engineering

University of Dhaka

Title: Implementation of the Bisection Method for finding roots of nonlinear equations

CSE 3212: Numerical Analysis Lab

Batch: 29/3rd Year 2nd Semester 2025

1 Objective(s)

- To understand the principle of the bisection method.
- To implement the iterative algorithm in a computer program.
- To approximate the root of a nonlinear function to a desired accuracy.
- To analyze the convergence and limitations of the method.

2 Theory

The Bisection Method is one of the simplest numerical techniques for finding a root of a nonlinear equation. It is also called the interval halving method. To use the Bisection method, one needs an initial interval that is known to contain a zero of the function. The method systematically reduces the interval. It does this by dividing the interval into two equal parts, performing a simple test, and based on the result of the test, half of the interval is thrown away. The procedure is repeated until the desired interval size is obtained.

If $f(x)$ is continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$, then there exists at least one root $c \in (a, b)$ such that $f(c) = 0$.

2.1 Procedure

1. Start with interval $[a, b]$ where the root exists.
2. Compute the midpoint:
$$c = \frac{a+b}{2}$$
3. Check the sign of $f(c)$:
   - If $f(a) \cdot f(c) < 0$, then the root lies in $[a, c]$.
   - Else if $f(c) \cdot f(b) < 0$, then the root lies in $[c, b]$.
4. Repeat until the interval length $|b - a|$ or $|f(c)|$ is less than the desired tolerance.

3 Algorithm

Input: function $f(x)$, interval $[a, b]$, tolerance ϵ, maximum iterations N.
Output: approximate root.

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Check if f(a) · f(b) < 0
if condition not satisfied then
    Stop (no root guaranteed in [a, b]).
end if
for k = 1 to N do
    c = (a + b)/2
    if f(c) = 0  or  |b − a| < ϵ then
        Stop: root ≈ c
    else if f(a) · f(c) < 0 then
        b = c
    else
        a = c
    end if
end for
Output c as approximate root.
```

4 Example Execution

We want to find the root of the function
$$f(x) = x^3 - x - 2$$
in the interval $[1, 2]$.

Step 1: Initial Interval

$$f(1) = 1^3 - 1 - 2 = -2, \quad f(2) = 2^3 - 2 - 2 = 4$$

Since $f(1) \cdot f(2) < 0$, the root lies in $[1, 2]$.

Step 2: Iterations

1. **Iteration 1:** Midpoint $c = \frac{1+2}{2} = 1.5$ $f(1.5) = (1.5)^3 - 1.5 - 2 = -0.125$ Sign check: $f(1) \cdot f(1.5) > 0$, so root lies in $[1.5, 2]$.
2. **Iteration 2:** $c = \frac{1.5+2}{2} = 1.75$ $f(1.75) = (1.75)^3 - 1.75 - 2 = 1.609$ Sign check: $f(1.5) \cdot f(1.75) < 0$, so root lies in $[1.5, 1.75]$.
3. **Iteration 3:** $c = \frac{1.5+1.75}{2} = 1.625$ $f(1.625) = (1.625)^3 - 1.625 - 2 = 0.666$ Root lies in $[1.5, 1.625]$.
4. **Iteration 4:** $c = \frac{1.5+1.625}{2} = 1.5625$ $f(1.5625) = (1.5625)^3 - 1.5625 - 2 = 0.252$ Root lies in $[1.5, 1.5625]$.
5. **Iteration 5:** $c = \frac{1.5+1.5625}{2} = 1.53125$ $f(1.53125) = (1.53125)^3 - 1.53125 - 2 = 0.0606$ Root lies in $[1.5, 1.53125]$.
6. **Iteration 6:** $c = \frac{1.5+1.53125}{2} = 1.515625$ $f(1.515625) = (1.515625)^3 - 1.515625 - 2 = -0.032$ Root lies in $[1.515625, 1.53125]$.

Step 3: Approximate Root

After several iterations, the interval narrows down to around $c \approx 1.5214$.

$$f(1.5214) \approx (1.5214)^3 - 1.5214 - 2 \approx 0$$

Therefore, the root is approximately $x = 1.5214$.

5 Lab Tasks (Please implement yourself and show the output to the instructor)

5.1 Problem 1

Consider the nonlinear function

$$f(x) = x^5 - 4x^3 + 2x^2 - \ln(x + 2) - 1.5$$

Use the **Bisection Method** to determine a real root of the equation $f(x) = 0$. The solution must be accurate to **at least five significant figures**. Use **six decimal places** in all intermediate calculations to avoid round-off errors. Employ initial guesses

$$x_\ell = 0.5, \qquad x_u = 1.5$$

After each iteration, report the following in tabular form:

- Lower bound x_ℓ
- Upper bound x_u
- Midpoint (approximate root) x_r
- Function value $f(x_r)$
- Approximate relative error $\epsilon_a(\%)$

Plot the absolute error versus the number of iterations to illustrate the convergence behavior of the bisection method.

5.2 Problem 2

In a microchannel, the flow depth y must satisfy the nonlinear equation

$$f(y) = \sqrt{\frac{2g}{L}} \left(\frac{A(y)}{P(y)} \right)^{3/2} - \frac{Q}{A(y)} = 0$$

where the cross-sectional area $A(y)$ and wetted perimeter $P(y)$ are given by

$$A(y) = 2y + y^2, \qquad P(y) = 2 + 2\sqrt{1 + y^2}$$

and the parameters are

$$Q = 1.2 \text{ m}^3/\text{s}, \quad g = 9.81 \text{ m/s}^2, \quad L = 15 \text{ m}.$$

Tasks:

1. Use the **Bisection Method** to compute the physically valid flow depth y.
2. Use the initial guesses
$$x_\ell = 0.3, \qquad x_u = 1.5.$$
3. Perform iterations until the stopping criterion
$$\epsilon_a \leq 0.5\% \quad \text{or} \quad n_{\max} = 12$$
is satisfied.
4. Show all iterations in tabular form.
5. Compute approximate relative error.
6. Plot a graph of absolute error vs the number of iterations to visualize the convergence of error.
7. Explain why only one root is physically meaningful.